

Password Manager CLI - User Manual

This document provides complete instructions for installing, running, and using the Password Manager CLI application.

1. Introduction This program is a command-line based password manager designed for personal use. It allows users to securely generate, store, update, delete, and decrypt passwords using encryption and a master password.

2. Requirements - Python installed (version 3.8 or higher) - Required package: cryptography Install using: `pip install cryptography`

3. Installation - Extract the project folder - Ensure the following structure exists:
`password_manager/` `main.py` `auth.py` `crypto.py` `generator.py` `manager.py` `data/` `anish.bat`

4. Running the Program - Open Command Prompt or PowerShell - Type: `anish` - Press Enter - On first run, you will be asked to set a master password

5. Authentication - The system uses a master password - Password is stored as a hash in `.auth` file - You must enter the correct password to access the program

6. Features Overview 1. Generate Password 2. List Stored Data 3. Modify Password 4. Delete Data 5. Reveal Passwords 6. Change Master Password 7. Exit

7. Generate Password - Enter platform name - Enter username - System generates strong password - Choose whether to save or not

8. List Feature - List all platforms and users - List specific platform - Shows encrypted passwords only

9. Modify Feature - Select platform and username - Generates new password - Updates stored value

10. Delete Feature - Delete user (platform + username) - Delete entire platform

11. Reveal Passwords - Requires master password again - Options: 1. View all passwords 2. View platform passwords 3. View single user password

12. File System - `.auth` : Stores hashed master password - `.key` : Encryption key - `.vault.json` : Stores encrypted passwords

13. Security Notes - Do not delete `.key` file - Do not share `.key` or `.vault.json` - Backup files regularly

14. Common Problems and Solutions Problem: `JSONDecodeError` Solution: Ensure `.vault.json` contains `{}` Problem: Invalid Fernet Key Solution: Delete `.key` and regenerate Problem: Permission Error Solution: Ensure program writes to correct directory Problem: Wrong Password Solution: Delete `.auth` and reset password

15. Technologies Used - Python - Cryptography (Fernet) - JSON for storage - OS module for file handling

16. Packages Used - cryptography

17. Notes This project is intended for personal and educational use. It demonstrates secure coding practices including hashing and encryption.

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